

Eligibility-Aware Hierarchical Task Alignment for Heterogeneous Respiratory Sound Learning

Abstract—Respiratory-sound datasets describe overlapping acoustic phenomena using different prediction units and annotation systems. Joint learning therefore requires an explicit account of which labels can supervise shared predictions. We study a shared BEATs model for ICBHI respiratory-cycle four-class classification and SPRSound event-level Normal/Adventitious classification. A three-node hierarchy combines Normal/Abnormal prediction with Crackle and Wheeze attributes; each node uses only annotation-supported targets, and separate readouts preserve the two native tasks. Across three runs, the model obtains an ICBHI Score of $61.17 \pm 0.31\%$ and a SPRSound official Score of $90.70 \pm 0.34\%$. Checkpoints are selected using ICBHI official-test Score, and these results are reported as a test-selected benchmark. The shared model produces both native outputs, while the ICBHI breakdown identifies Wheeze and Both as the least accurately recognized classes.

Index Terms—respiratory sounds, joint learning, heterogeneous annotations, hierarchical classification, label availability

I. INTRODUCTION

Respiratory-sound learning draws on datasets that describe related acoustic phenomena at different levels. ICBHI labels complete respiratory cycles as Normal, Crackle, Wheeze, or Both, whereas SPRSound provides event annotations and an event-level Normal/Adventitious task [1], [2]. A model shared across these datasets must respect both the meaning of an annotation and the prediction that each benchmark requests. In particular, an annotation that supports an abnormality decision need not specify whether Crackle or Wheeze is present.

Strong performance within one source task is not sufficient evidence for reusing its fixed decision rule on another benchmark. Existing ICBHI-specialized checkpoints provide a concrete motivation for examining this issue on SPRSound, as reported in Section IV. Such fixed-head transfer measures direct model reuse; it does not determine whether the encoder’s features would support a newly trained target classifier.

Related work already addresses substantial parts of heterogeneous learning. The SPRSound study includes an ICBHI/SPRSound data-fusion benchmark with a common label space [2]. LungMix aligns respiratory-sound labels to a common four-class system for single-source domain generalization [3]; OPERA learns shared respiratory representations through self-supervised pretraining and evaluates task-specific downstream predictors [4]. Partial-label methods handle overlapping label spaces and missing annotations [5], [6]. In audio, DCASE 2024 Task 4 combines heterogeneous annotations, label mapping, and masking of unavailable targets [7]. Attribute-based respiratory classification also has precedents, including PC-MCL [8]. These advances motivate examining

how annotation-supported attributes can be shared between respiratory cycles and events with different native readouts.

We study their use in joint ICBHI cycle-level four-class and SPRSound event-level binary learning. Our formulation expresses compatible supervision through shared Normal/Abnormal, Crackle, and Wheeze nodes, while retaining separate native readouts. We report a three-run benchmark, fixed-checkpoint transfer context, and class-specific behavior. The analysis examines joint native-task prediction together with category-specific recognition, keeping each dataset’s prediction unit and evaluation metrics explicit.

II. DATA AND TASK ALIGNMENT

A. Native Units and Evaluation Tasks

ICBHI supplies annotated respiratory cycles with four labels: Normal, Crackle, Wheeze, and Both [1]. We use its official recording split, comprising 4,142 training and 2,756 test cycles; this split is not a strict patient-held-out test. The test supports are 1,579 Normal, 649 Crackle, 385 Wheeze, and 143 Both cycles. SPRSound supplies respiratory events with more detailed sound labels [2]. Our primary SPRSound endpoint is BioCAS2022 Task1-1 on the official inter-subject test: 1,429 events, including 1,040 Normal and 389 Adventitious events. Its intra-subject partition and seven-class task are outside this primary endpoint.

The training portions provide fitting and group-disjoint internal validation samples. For ICBHI, the first fold of a shuffled, five-fold stratified group split defines validation, with patient identifiers as groups and the run seed controlling the split. The 6,656 official SPRSound training events are split into 5,219 fitting and 1,437 validation events, with 194 and 49 patient groups, respectively. These internal data are used to set the shared attribute thresholds; checkpoint selection for the reported benchmark is specified separately in Section IV.

B. Annotation-Supported Targets

Let a , c , and w denote abnormality, Crackle, and Wheeze targets. ICBHI labels determine all three. Compatible SPRSound labels use the same mapping, with Coarse and Fine Crackle both supporting the Crackle attribute. Rhonchi and Stridor supervise abnormality but leave the two attributes unavailable. Thus, the binary SPRSound evaluation endpoint does not imply binary-only training: supported raw event labels also supervise the attributes. An unavailable target contributes neither positive nor negative classification supervision. Figure 1 summarizes this mapping; the native unit remains a cycle or an event.

Native annotation	a	c	w
Normal (ICBHI / SPRSound)	0	0	0
Crackle (ICBHI) / Coarse or Fine (SPR)	1	1	0
Wheeze (ICBHI / SPRSound)	1	0	1
Both (ICBHI) / Wheeze+Crackle (SPR)	1	1	1
Rhonchi or Stridor (SPRSound)	1	–	–

Fig. 1. Shared targets from native annotations. A dash denotes an unavailable attribute, excluded from its classification loss. ICBHI retains cycle-level four-class prediction; SPRSound retains event-level Normal/Adventitious prediction.

C. Input Construction

Each recording is converted to mono and resampled to 16 kHz. Following the inherited input pipeline, a linear fade of 1/16 s is applied at recording boundaries before extracting the annotated native unit. Units longer than 5 s retain their first 5 s; shorter units are repeated to 5 s and receive an end fade. Thus each cycle or event supplies one fixed-length model input, and one native prediction, without introducing additional window-level labels.

III. METHOD

A. Shared Representation and Prediction Nodes

Figure 2 shows the model and supervision flow. We initialize the encoder with BEATs iter3+ AS2M [9] and fine-tune it on both core datasets. Its filterbank frontend and Transformer produce a sequence of 768-dimensional features. Mean pooling over this sequence, followed by a shared linear projection to 256 dimensions, supplies three linear predictors: a two-logit Normal/Abnormal head and two scalar attribute heads. For pooled features h , their outputs are

$$\begin{aligned} p_a &= \text{softmax}(f_a(h)), \\ p_c &= \sigma(f_c(h)), \quad p_w = \sigma(f_w(h)). \end{aligned} \quad (1)$$

The attributes are marginal sigmoid predictions. The Normal/Abnormal decision gates ICBHI decoding; it is not multiplied into the attribute probabilities.

B. Eligibility-Aware Classification Objective

For sample i and node $k \in \{a, c, w\}$, let $m_{ik} = 1$ when the source annotation supports target y_{ik} , and 0 otherwise. In a batch B , define $B_k = \{i \in B : m_{ik} = 1\}$ and $K_B = \{k : |B_k| > 0\}$. The classification loss is

$$\mathcal{L}_{\text{cls}}(B) = \frac{1}{|K_B|} \sum_{k \in K_B} \frac{1}{|B_k|} \sum_{i \in B_k} \ell_k(p_{ik}, y_{ik}), \quad (2)$$

where ℓ_a is two-class cross-entropy and ℓ_c, ℓ_w are binary cross-entropy. Normalization occurs separately within each available node. Dataset-homogeneous batches preserve native units, and both datasets supply an equal number of batches per epoch.

We inherit PAFA’s patient cohesion-separation and global patient alignment objectives [10]. An auxiliary attention-based projection of the encoder sequence supplies their training

features; patient identifiers are qualified by dataset. With the inherited weights, the total objective is

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + 50\mathcal{L}_{\text{PCSL}} + 0.0005\mathcal{L}_{\text{GPAL}}. \quad (3)$$

This auxiliary projection does not enter the native classification readouts.

C. Thresholds and Native Readouts

The Level-1 argmax gives the Normal/Abnormal decision. Shared Crackle and Wheeze thresholds are chosen independently from internal validation predictions by maximizing the equal-dataset mean of the eligible-node binary F1 scores; exact ties select the larger threshold. The candidates are the observed eligible probabilities together with 0 and 1. At each epoch, thresholds are fixed before the official ICBHI readout.

For ICBHI, a Normal Level-1 decision produces Normal. An Abnormal decision produces Both when both attributes exceed their thresholds, and the corresponding class when exactly one does. If neither does, the larger probability-minus-threshold margin selects Crackle or Wheeze, with ties assigned to Crackle. SPRSound uses only the Level-1 decision for its native Normal/Adventitious output. Neither task is evaluated through a pooled score.

IV. EVALUATION

A. Experimental Setup and Selection

The joint benchmark uses full BEATs fine-tuning, batches of 32, Adam with learning rate 5×10^{-5} and weight decay 10^{-6} , a 50-epoch cosine schedule with minimum learning-rate ratio 0.001, and the inherited parameter EMA coefficient 0.5. Each epoch contains 326 updates, equally divided between the datasets. SpecAugment, mean-variance normalization, and HF auxiliary supervision are disabled. The three formal seeds are 0, 1, and a fresh 42.

Checkpoint selection and patience-10 early stopping use the ICBHI official four-class test Score. Strict improvements replace the selected checkpoint; ties retain the earlier epoch. Selected epochs are 15, 11, and 17 after 25, 21, and 27 completed epochs, respectively. Each selected checkpoint is then evaluated once on SPRSound official inter. The reported mean and sample standard deviation therefore describe an ICBHI-test-selected benchmark, following the selection convention of the PAFA reference implementation. They are not estimates from a clean validation-selected protocol.

B. Metrics and Comparison Sources

For ICBHI, Sp is the fraction of Normal cycles correctly classified, while Se is the fraction of abnormal cycles assigned to their correct abnormal subclass; its Score is $(Sp + Se)/2$. Confusing Crackle with Wheeze reduces Se even though the cycle is recognized as abnormal. For SPRSound Task1-1, Se and Sp are Adventitious and Normal recall, respectively. Its arithmetic and harmonic scores are $AS = (Se + Sp)/2$ and $HS = 2SeSp/(Se + Sp)$, with official Score $(AS + HS)/2$.

Table I separates published ICBHI references, local fixed-checkpoint evaluations, and the joint benchmark. Published

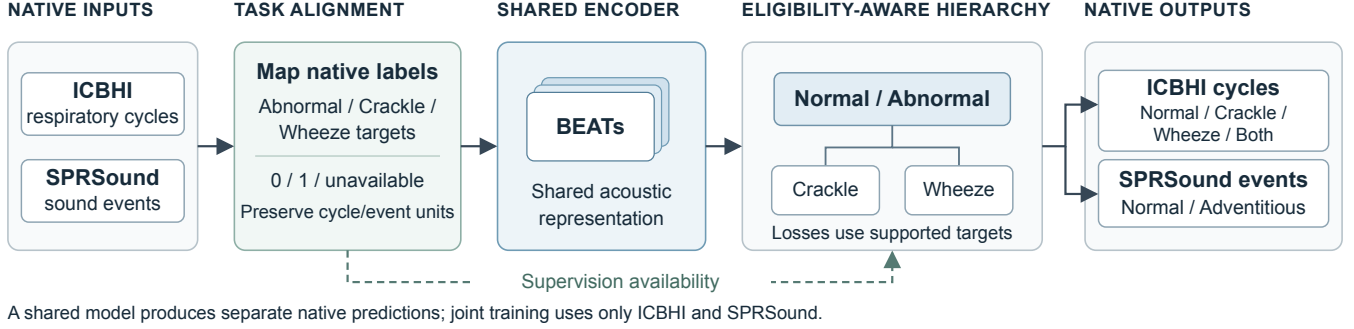


Fig. 2. Joint learning with native task readouts. Native cycles and events retain their prediction units. Task alignment supplies shared targets and node availability; a shared BEATs encoder feeds Normal/Abnormal, Crackle, and Wheeze predictors. Unavailable targets are omitted from the corresponding node loss. ICBHI uses hierarchical four-class decoding, while SPRSound uses the Normal/Abnormal output. The dashed arrow denotes supervision availability, not a second acoustic input. HF/KAUH are external diagnostics and are not training inputs to this model.

TABLE I

NATIVE-TASK BENCHMARK AND FIXED-CHECKPOINT TRANSFER CONTEXT, IN PERCENT. PUBLISHED ROWS RETAIN THE SOURCE MEAN $\pm$ STANDARD DEVIATION; THE JOINT ROW USES SAMPLE STANDARD DEVIATION, AND LOCAL ROWS USE ONE SOURCE CHECKPOINT EACH. ALL TRAINED ROWS SHOWN USE ICBHI-TEST-SELECTED SOURCE CHECKPOINTS OR THE CORRESPONDING PUBLISHED SELECTION CONVENTION. DASHES DENOTE UNREPORTED METRICS. SPRSOUND TRANSFER VALUES ARE AS, NOT OFFICIAL SCORE; THE ALL-NORMAL 50.00 REFERENCE APPLIES ONLY TO AS.

Model / evidence source	ICBHI: cycle four-class			SPRSound: event binary	
	Sp	Se	Score	AS	Official Score
<i>A. Published ICBHI context: five runs</i>					
BEATs+CE [10]	78.77 $\pm$ 3.07	48.21 $\pm$ 2.32	63.49 $\pm$ 1.08	—	—
BEATs+PAFA [10]	82.05 $\pm$ 1.95	47.63 $\pm$ 2.23	64.84 $\pm$ 0.60	—	—
<i>B. Local source checkpoints and zero-target transfer: one checkpoint per method</i>					
PAFA	76.88	51.40	64.14	55.82	—
SG-SCL (device-aware source)	74.98	46.98	60.98	59.98	—
Patch-Mix	86.19	38.15	62.17	59.38	—
All-Normal target reference	—	—	—	50.00	—
<i>C. Joint hierarchy: three formal runs (0, 1, 42)</i>					
Shared BEATs hierarchy	74.98 $\pm$ 5.69	47.35 $\pm$ 5.35	61.17 $\pm$ 0.31	90.95 $\pm$ 0.31	90.70 $\pm$ 0.34

PAFA rows are five-run results [10]; their values are not paired with local transfer scores as though they came from the same checkpoint. The local PAFA, SG-SCL, and Patch-Mix rows each describe one fixed source checkpoint [10]–[12]. SG-SCL source training uses device metadata. Their SPRSound evaluations collapse source four-class argmax predictions to Normal versus the remaining classes, without target training, calibration, or threshold search. These source recipes and decoders differ from the joint model and are contextual references, not matched ablations.

C. Current Native-Task Results

The same selected joint model supports both native tasks, obtaining an ICBHI Score of $61.17 \pm 0.31\%$ and a SPRSound official Score of $90.70 \pm 0.34\%$. Its SPRSound AS is separately $90.95 \pm 0.31\%$. The fixed expert checkpoints obtain SPRSound AS values of 55.82–59.98%, above the all-Normal AS reference but limited under their direct-reuse conditions. This comparison does not isolate the gain from joint training: the joint model receives SPRSound supervision, whereas the expert transfers do not. It also does not test the transferability of frozen features with a new target-trained head.

ICBHI aggregate variation is small relative to the variation of its components: Score standard deviation is 0.31 percentage points, compared with 5.69 for Sp and 5.35 for Se . Per-class recall is $74.98 \pm 5.69\%$ for Normal, $60.20 \pm 8.48\%$ for Crackle, $29.61 \pm 2.08\%$ for Wheeze, and $36.83 \pm 11.37\%$ for Both. These are descriptive differences across the three runs, not a significance claim or a causal explanation. Relative to the local PAFA checkpoint, both Sp and Se are lower; the difference cannot be described as a cost confined to specificity. SPRSound Normal and Adventitious recall are $84.29 \pm 1.36\%$ and $97.60 \pm 1.21\%$, respectively.

D. Required Controls and Supporting Diagnostics

Table II identifies the comparisons needed to assess joint supervision and hierarchical alignment. None has a completed result under the proposed common validation-selected protocol. A group-disjoint calibration/selection design, a common joint validation criterion, and matched source-only exposure remain to be specified. The proposed independent SPRSound head uses binary targets, whereas the hierarchy also uses supported event attributes. This comparison changes supervision as well as classifier structure; isolating hierarchy alone would

TABLE II
PROSPECTIVE MATCHED CONTROLS. ALL FOUR RESULTS ARE
CURRENTLY UNAVAILABLE UNDER THE INTENDED
VALIDATION-SELECTED PROTOCOL. THE EXISTING
ICBHI-TEST-SELECTED JOINT ROW IN TABLE I IS NOT A SUBSTITUTE.

Training condition	Required comparison
Joint hierarchy	Validation-selected reference
Joint independent native heads	Joint design comparison
ICBHI-only hierarchy	Joint vs. source-only learning
SPRSound-only hierarchy	Reverse source-only comparison

require matching that supervision. The source-only conditions also require fixed-checkpoint transfer to the other native task without target calibration or selection. The current benchmark cannot establish relative superiority or quantify joint-training effects.

Fixed selected checkpoints also provide bounded post-hoc diagnostics on HF Lung [13] and KAUH [14]. HF positive-interval recall is 0.5828 ± 0.1263 for Crackle-compatible annotations and 0.8543 ± 0.0107 for Wheeze. An interval is counted as recognized when any overlapping 5-s window exceeds the fixed attribute threshold. Unannotated gaps are not negatives, and these window-overlap measurements do not establish precise temporal localization. KAUH patient-level binary and compatible four-class average scores are 0.7217 ± 0.0164 and 0.4700 ± 0.0830 ; B/D/E recordings are pooled within each patient, with unresolved labels excluded. These different readout tasks are not native reproductions or evidence of comparable performance across four datasets.

V. CONCLUSION

A shared respiratory-sound model can use annotation-supported supervision while retaining ICBHI cycle-level four-class and SPRSound event-level binary outputs. The three-run test-selected benchmark demonstrates joint prediction capability alongside uneven class-specific recognition, particularly for ICBHI Wheeze and Both. Matched independent-head and single-dataset comparisons remain necessary to establish the additional value of the joint alignment design.

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